

WENDA XU

Research Scientist at Google Deepmind (Working on Large Language Model Evaluation and Post-training, Google LLC)

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Webpage

Github

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RESEARCH INTERESTS

My research focuses on improving large language models (LLMs) through rigorous evaluation and efficient post-training. At Gemini, I currently lead the evaluation efforts for speech translation and work on RL post-training for streaming speech translation.

Previously, I developed automated methods to assess model capabilities, including efficient data curation techniques for building challenge benchmarks [2] [3] [4]. I also specialize in designing explainable and learned evaluation metrics. This research led to the development of SEScore1&2 [13][14] and InstructScore [11][12], the latter of which was recognized as the best unsupervised metric at the WMT22 shared task. Furthermore, my background includes developing efficient post-training techniques ranging from inference time optimization [9], preference learning with BPO [7], to knowledge distillation via Speculative KD [6].

INDUSTRY RESEARCH EXPERIENCE

Full-time Research Scientist

Google Deepmind (Google LLC)

April 2025 – Now

Mountain View, CA

- Lead speech translation evaluation effort at Gemini.
- Actively work on RL post-training on streaming translation at Gemini audio.
- Developed and scaled an automated pipeline to generate dynamic challenge sets for evaluating Gemini models across both pre-training and post-training stages [2].
- Led the research efforts in automatic benchmark construction using LLMs, studying both self-bias in LLM-generated benchmark and efficient auto-challenge set construction [3][4][8].
- Led the research efforts in studying length bias of translation evaluation metric [5].
- **Product Impact:** [Live Translate Feature at Google Translate](#), streaming translation at Google Meet, Youtube dubbing, Sports commentary etc.
- **Publication:** [TranslateGemma Technical Report](#)
- **Publication:** [Searching for Difficult-to-Translate Test Examples at Scale](#)
- **Publication:** [Deconstructing Self-Bias in LLM-generated Translation Benchmarks](#)
- **Publication:** [Generating Difficult-to-Translate Texts](#)
- **Publication:** [Penalizing Length: Uncovering Systematic Bias in Quality Estimation Metrics](#)

Student Researcher Intern

Google Cloud Research (Google LLC)

Jun 2024 – Oct 2024

Los Angeles, CA

- Mentors: Rujun Han, Zifeng Wang, Rishabh Agarwal, Chen-Yu Lee.
- **Publication:** [Speculative Knowledge Distillation: Bridging the Teacher-Student Gap Through Interleaved Sampling](#)
- Built a generic KD framework that generalizes to on-policy and supervised KD, achieves substantial gains in task specific and task agnostic knowledge distillation
- **Product Impact:** Deployed in production at Google Translate.

Student Researcher Intern

Google Translate Research (Google LLC)

Jun 2023 – Dec 2023

Mountain View, CA

- Mentors: Dan Deutsch, Markus Freitag.
- **Publication:** [LLMRefine: Pinpointing and Refining Large Language Models via Fine-Grained Actionable Feedback](#)
- Developed an efficient, inference-time optimization technique that iteratively refines PaLM 2 outputs at the span level,
- **Product Impact:** Deployed in production at the YouTube team.

Research Science Intern

ByteDance AI Lab

📅 Jun 2022 – Oct 2022

📍 Mountain View, CA

- Mentors: Xian Qian, Mingxuan Wang.
- **Publication:** [SESCORE2: Learning Text Generation Evaluation via Synthesizing Realistic Mistakes](#)
- Developed a learned evaluation metric without human labels that achieved high correlation with human judgment and greatly improves the assessment of translation quality.

EDUCATION

Ph.D., Computer Science (GPA: 3.8/4.0)

University of California, Santa Barbara

📅 Sept 2020 – Mar 2025

📍 Santa Barbara, CA

- Dissertation: [On Evaluation and Efficient Post-training for LLMs](#)
- Advisors: William Yang Wang, Ph.D., Lei Li, Ph.D.

B.S., Computer Science (GPA: 3.9/4.0)

University of California, Davis

📅 Sept 2016 – Mar 2020

📍 Davis, CA

- Senior Design Project: Visual SLAM using ORB-SLAM2 with Path Finding (**Best Senior Design of a Year**)
- Advisors: Chen-Nee Chuah, Ph.D.

PATENTS

Knowledge Distillation with Interleaved Sampling [6] was filed a patent with legacy IDF number IDF-320078.

SELECTED PUBLICATIONS

A full list of my publications is available on my [Google Scholar profile](#).

- [1] **Wenda Xu as a contributor**, Translate Research Team. [TranslateGemma](#). Preprint. Official Gemma release: Optimized for translation related tasks with SOTA performance by size.
- [2] **Wenda Xu***, Vilém Zouhar*, Parker Riley, Mara Finkelstein, Markus Freitag, Daniel Deutsch. [Searching for Difficult-to-Translate Test Examples at Scale](#). Preprint. A scalable and efficient automatic data curation technique for building challenge benchmarks by finding the most difficult test examples.
- [3] **Wenda Xu**, Sweta Agrawal, Vilém Zouhar, Markus Freitag, Daniel Deutsch. [Deconstructing Self-Bias in LLM-generated Translation Benchmarks](#). Preprint. We demonstrate that LLM-generated benchmarks exhibit a systematic self-bias that inflates a model's own performance, and low source text diversity is one primary cause.
- [4] Vilém Zouhar, **Wenda Xu**, Parker Riley, Juraj Juraska, Mara Finkelstein, Markus Freitag, Daniel Deutsch. [Generating Difficult-to-Translate Texts](#). Preprint. MT-breaker, employs a LLM to iteratively generate difficult yet natural prompt examples to build more challenging translation benchmarks.
- [5] Yilin Zhang, **Wenda Xu**, Zhongtao Liu, Tetsuji Nakagawa, Markus Freitag. [Penalizing Length: Uncovering Systematic Bias in Quality Estimation Metrics](#). Preprint. Systematic Study of Length Bias in Quality Estimation Metrics at Translation.
- [6] **Wenda Xu**, Rujun Han, Zifeng Wang, Long Le, Dhruv Madeka, Lei Li, William Yang Wang, Rishabh Agarwal, Chen-Yu Lee, Tomas Pfister. [Speculative Knowledge Distillation: Bridging the Teacher-Student Gap Through Interleaved Sampling](#). ICLR 2025. A generic KD framework that generalizes to on-policy and supervised KD, achieves substantial gains in task specific and task agnostic knowledge distillation.

- [7] **Wenda Xu***, Jiachen Li*, William Yang Wang, Lei Li. *BPO: Supercharging Online Preference Learning by Adhering to the Proximity of Behavior LLM*. EMNLP 2024 (*equal contribution). On-policy BPO achieves superior performance compared to DPO on TL;DR (89.5% vs 72.0%), Helpfulness (93.5% vs 82.2%), and Harmfulness (97.7% vs 77.5%).
- [8] **Wenda Xu**, Guanglei Zhu, Xuandong Zhao, Liangming Pan, Lei Li, William Yang Wang. *Pride and Prejudice: LLM Amplifies Self-Bias in Self-Refinement*. ACL 2024 Oral. The study to define and quantify the bias exhibited by LLMs when assessing their own generated outputs.
- [9] **Wenda Xu**, Daniel Deutsch, Mara Finkelstein, Juraj Juraska, Biao Zhang, Zhongtao Liu, William Yang Wang, Lei Li, Markus Freitag. *LLMRefine: Pinpointing and Refining Large Language Models via Fine-Grained Actionable Feedback*. NAACL 2024. An efficient inference time optimization technique that iteratively refines the outputs of the PaLM 2 model at text span-level, achieving improvements of 1.7 MetricX on translation, 8.1 ROUGE-L on ASQA, and 2.2 ROUGE-L on topical summarization.
- [10] Liangming Pan, Michael Saxon, **Wenda Xu**, Deepak Nathani, Xinyi Wang, William Yang Wang. *Automatically Correcting Large Language Models: Surveying the landscape of diverse self-correction strategies*. TACL 2024. This work provides a summary and analysis of self-correction techniques.
- [11] **Wenda Xu**, Danqing Wang, Liangming Pan, Zhenqiao Song, Markus Freitag, William Yang Wang, Lei Li. *INSTRUCTSCORE: Explainable Text Generation Evaluation with Finegrained Feedback*. EMNLP 2023 Oral. A fine-grained 7B LLM autorater, trained on synthetic data, surpasses unsupervised metrics, including those based on 175B GPT-3 and GPT-4.
- [12] Chinmay Dandekar, **Wenda Xu**, Xi Xu, Siqi Ouyang, Lei Li. *Translation Canvas: An Explainable Interface to Pinpoint and Analyze Translation Systems*. EMNLP Demo 2024. System demonstration of InstructScore.
- [13] **Wenda Xu**, Xian Qian, Mingxuan Wang, Lei Li, William Yang Wang. *SEScore2: Learning Text Generation Evaluation via Synthesizing Realistic Mistakes*. ACL 2023. Kendall correlation improved 14.3% from SEScore.
- [14] **Wenda Xu**, Yilin Tuan, Yujie Lu, Michael Saxon, Lei Li, William Yang Wang. *Not All Errors are Equal: Learning Text Generation Metrics using Stratified Error Synthesis*. EMNLP 2022. SEScore: No.1 unsupervised metric at WMT22 metrics shared task.
- [15] **Wenda Xu**, Michael Saxon, Misha Sra, William Yang Wang. *Self-Supervised Knowledge Assimilation for Expert-Layman Text Style Transfer*. AAAI 2022. Relative improvement in overall success rate at text style transfer by 106%.

COLLABORATIVE PUBLICATIONS

- [1] Xi Xu, **Wenda Xu**, Siqi Ouyang, Lei Li. *CA*: Addressing Evaluation Pitfalls in Computation-Aware Latency for Simultaneous Speech Translation*. NAACL 2025
- [2] Juhyun Oh, Eunsu Kim, Jiseon Kim, **Wenda Xu**, Inha Cha, William Yang Wang, Alice Oh. *Uncovering Factor Level Preferences to Improve Human-Model Alignment*. EMNLP 2025
- [3] Michael Saxon, Xinyi Wang, **Wenda Xu**, William Yang Wang. *PECO: Examining Single Sentence Label Leakage in Natural Language Inference Datasets through Progressive Evaluation of Cluster Outliers*. EACL 2023
- [4] Yujie Lu, Weixi Feng, Wanrong Zhu, **Wenda Xu**, Xin Eric Wang, Miguel Eckstein, William Yang Wang. *Neuro-Symbolic Causal Language Planning with Commonsense Prompting*. ICLR 2023
- [5] Wanrong Zhu, An Yan, Yujie Lu, **Wenda Xu**, Xin Eric Wang, Miguel Eckstein, William Yang Wang. *Visualize Before You Write: Imagination-Guided Open-Ended Text Generation*. EACL 2023

[6] Yi-Lin Tuan, Alon Albalak, **Wenda Xu**, Michael Saxon, Connor Pryor, Lise Getoor, William Yang Wang.
CausalDialogue: Modeling Utterance-level Causality in Conversations. ACL 2023

SKILLS

Software Proficiencies

Python

Pytorch

Numpy

C

C++

Linux



Conceptual

Natural Language Processing

Large Language model (LLM)

LLM Evaluation

Post-training



HONORS

- The Robert Noyce Fellowship, Academic Excellence Fellowship (UC Santa Barbara, 2022)
- Honor Graduation (UC Davis, 2020)
- Thomas E. Bruzzone Award (UC Davis, 2019)
- Robert Murdoch Memorial Scholarship (UC Davis, 2019)
- Best Senior Design of a Year (Visual SLAM) (UC Davis, 2019)
- College of Engineering Dean’s Honor list (UC Davis, 2016-2020)